

CurbRisk-AV — Quantifying autonomous vehicle pickup and drop-off risk using exposure-adjusted complaint data

Final Paper

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Paper Under Construction

Planned completion by ...

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Abstract

TODO: abstract goes here (write last).

1 Introduction

1.1 Problem Statement

1.1.1 The Problem:

Autonomous-vehicle pick-up / drop-off (PUDO) maneuvers — the moments a driverless vehicle stops so a rider can get in or out — are a plausible source of curb-space conflict, minor collisions, and rider complaints. Whether that risk is growing, shrinking, or holding steady as a fleet scales is hard to answer from the public record: California Public Utilities Commission (CPUC) AV Deployment reports are large, redacted, and change format from quarter to quarter, so nobody has simply lined them up into a rate over time.

Beyond Waymo specifically, PUDO-type friction is a useful early empirical signal for autonomous-vehicle deployment more broadly: as AV operations expand to new cities, road layouts, and weather conditions, understanding how curb-side risk behaves under a scaling single-carrier deployment is a natural first step. The rates found here, however, come from one carrier operating in California and should not be read as representative of AV operations generally (see Scope, below).

1.1.2 Contribution

- A reproducible pipeline (**pudo-pipeline**) that ingests nine CPUC quarterly report zips for Waymo’s driverless deployment — two incompatible file schemas, redaction, and formatting drift included — into five analysis-ready tables.
- An exposure-adjusted PUDO complaint rate across eight quarters (2024 Q3 – 2026 Q2), on trips — the team’s chosen exposure measure, cross-checked against total vehicle miles traveled — with an explicit primary trend test (Negative-Binomial GLM, after Poisson regression was diagnosed as overdispersed), a launch-period sensitivity analysis isolating the 2024 deployment ramp, a distribution-free cross-check (Mann-Kendall), and a Bayesian refit (PyMC/NUTS) that reports a direct posterior probability the rate is declining alongside the frequentist p-value.
- A placement of the PUDO hazard on a probability $\times$ impact risk matrix that is computed from the data rather than asserted, with the probability basis left as a visible, toggleable assumption.

1.1.3 Scope and Vision

Scope: one carrier (Waymo), one state (California), eight quarters, public CPUC filings only — no proprietary telemetry. **Vision:** the pipeline’s loaders are written against the CPUC’s public template, not against Waymo specifically, so the same numerator/denominator/trend

method can be pointed at any other CPUC-reporting AV carrier to support carrier-to-carrier comparison rather than a single-carrier trend.

1.2 Technical Challenges

- **Two incompatible numerator schemas.** 2024 reports ship one aggregate, pre-summed row per quarter; 2025Q1 onward ship ride-level microdata (millions of rows) with a Y/N complaint flag. Both had to be reconciled into one numerator, and the level shift between eras cannot be fully ruled out.
- **Redaction.** Incident timestamps, locations, and the `PUDOTravelLane` flag are blank in every file of every period, ruling out within-quarter timing or spatial analysis.
- **Silent formatting drift.** Comma-grouped numerics, filenames that change across quarters, and a transposed carrier ID silently nulled the exposure data until caught — the loader now warns loudly instead of nulling quietly.
- **Non-calendar reporting periods.** The 2024 CPUC reports are not calendar quarters, so “2024 Q3” / “2024 Q4” needed explicit relabeling (`period_label`, `is_calendar_quarter`) to avoid silently comparing the wrong months.
- **Small, overdispersed counts.** 27–79 complaints per quarter is enough to rule out the normal-approximation trend test, but not enough to ignore quarter-to-quarter noise.

2 Related Work

3 Data

3.1 Data Sources

Public data comes from the CPUC AV Program’s quarterly deployment reports (cpuc.ca.gov → Licensing → Autonomous Vehicle Programs → Quarterly Reporting) [1]. Two file families, both filed per reporting period:

- **Incidents-Complaints** — the numerator. PUDO complaint and PUDO-collision counts.
- **Month-Level** — the denominator. Real `Year/Month`, trips, and vehicle miles traveled (VMT). VMT is further split into three “period” columns in the source file, but CPUC’s own reference key never defines what those periods mean, so this analysis uses trips as the primary exposure measure and VMT only as an unsplit, whole-quarter cross-check.

Nine raw reporting windows (Jun–Aug 2024, Sep–Nov 2024, a Dec 2024 one-month stub, then eight calendar quarters through 2026 Q2) are recombined into eight analysis quarters, 2024Q3–2026Q2.

The pipeline’s two upstream file families reduce to a single hand-off table, `final_pudo_summary` — one row per analysis quarter, carrying only the complaint numerator and the trips denominator (Table 1). This is the dataset the rest of the paper’s analysis runs on:

Table 1: The final hand-off dataset (`final_pudo_summary`): PUDO complaint counts and trip exposure by analysis quarter, 2024 Q3 - 2026 Q2 ($n = 8$).

period	trips	PUDO complaints
2024 Q3 (Jun-Aug)	751,844	41
2024 Q4 (Sep-Dec)	1,852,614	51
2025 Q1	1,818,206	27
2025 Q2	2,234,912	42
2025 Q3	2,676,001	42
2025 Q4	3,748,532	66
2026 Q1	3,905,934	69
2026 Q2	4,220,075	79

All numbers, tables, and figures in this paper are pulled directly from the project’s analysis notebook — see [pudo_analysis.ipynb](#) for the full code and every intermediate result.

4 Methods

4.1 Feature Engineering

Every analysis period carries a clean sort/filter key (`period_id`) alongside a human-readable label, its exact date window, and a flag for whether that window is actually a calendar quarter (`period_label`, `window_start/window_end`, `is_calendar_quarter` — false only for the two 2024 periods). Two exposure denominators are computed per period — trips (one pick-up + one drop-off per trip, the most literal PUDO-opportunity count; the primary measure) and total VMT (a cross-check) — each paired with a `pudo_per_100k_*` rate, plus a PUDO-share-of-all-complaints column for context.

4.2 Modeling

PUDO complaints were initially modeled using Poisson regression with the logarithm of trip exposure included as an offset. A variance-to-mean dispersion diagnostic (Equation 3) showed substantial overdispersion in the quarterly counts ($\text{VMR} \approx 5.9$), violating the Poisson equidispersion assumption. Temporal trends were therefore estimated using **Negative-Binomial regression** with a log link and a trip-exposure offset as the primary model. Because only eight quarterly observations were available, the same model was additionally estimated in a **Bayesian framework** to more fully characterize parameter and dispersion uncertainty. The Poisson fit below is retained only as an initial diagnostic baseline — it is never reported as the trend result.

4.2.1 Distributional diagnostics

Before any count model is fit, the raw quarterly PUDO rate is characterized directly: histograms (mean/median annotated), normal Q-Q plots, and a Shapiro-Wilk test (Table 5;

H_0 : the rate values are drawn from a normal distribution) are computed for both the full window and the post-ramp window introduced below. Two two-sample Kolmogorov-Smirnov (KS) comparisons are also run, but only where a real, adequately-sized empirical sample exists on both sides: the 2024 launch-quarter rates against the 2025–2026 rates (descriptive only — $n = 2$ vs. $n = 6$ is under-powered), and the monthly trip-exposure distributions for 2025 ($n = 12$) against 2026 ($n = 6$, partial year), which is the one case here where a two-sample KS test is genuinely well-powered. These diagnostics motivate, rather than assume, the choice of a count-model (Poisson/Negative-Binomial) framework over classic parametric mean-comparison tests for the complaint rate throughout the rest of this paper.

4.2.2 Poisson regression (initial diagnostic baseline)

The complaint count Y_t in analysis quarter $t = 0, \dots, 7$ is first modeled as a Poisson process whose rate scales with exposure E_t (trips, or VMT as a cross-check), with a log-linear time trend:

$$Y_t \mid \mu_t \sim \text{Poisson}(\mu_t), \quad \log \mu_t = \log E_t + \beta_0 + \beta_1 t \quad (1)$$

$\log E_t$ enters with a fixed coefficient of 1 (an *offset*), so β_1 is the log change in the underlying **rate** μ_t/E_t per quarter, not in the raw count. Every incidence rate ratio (IRR) and “per-quarter % change” quoted in this paper, for whichever model produced β_1 , is:

$$\text{IRR} = e^{\beta_1}, \quad \% \text{ change per quarter} = (e^{\beta_1} - 1) \times 100\% \quad (2)$$

4.2.3 Overdispersion check

Equation 1’s defining assumption is $\text{Var}(Y_t) = \mathbb{E}[Y_t]$. This is checked with the index of dispersion (variance-to-mean ratio, VMR) on the raw complaint counts across the $n = 8$ quarters:

$$\text{VMR} = \frac{s^2}{\bar{y}}, \quad D = (n - 1) \text{VMR} \sim \chi_{n-1}^2 \quad \text{under } H_0 : \text{Poisson} \quad (3)$$

$\text{VMR} \approx 1$ is consistent with a pure Poisson process; $\text{VMR} \gg 1$ — the result found here ($\text{VMR} = 5.9$) — rejects equidispersion, so the Poisson fit’s confidence interval and p -value understate uncertainty and are not used for inference about the trend.

4.2.4 Negative-Binomial regression (primary trend model)

Because Equation 3 rejects equidispersion, the trend is instead estimated with Negative-Binomial regression (the NB2 parametrization used by `statsmodels` [2]), keeping the same log-linear mean structure and offset as Equation 1:

$$Y_t \mid \mu_t, \alpha \sim \text{NB}(\mu_t, \alpha), \quad \text{Var}(Y_t) = \mu_t + \alpha \mu_t^2, \quad \log \mu_t = \log E_t + \beta_0 + \beta_1 t \quad (4)$$

with the dispersion parameter α estimated by maximum likelihood alongside β_0, β_1 (larger α means *more* overdispersion relative to Poisson under this parametrization). This is the paper’s **primary frequentist trend model**, reported with β_1 , its standard error, 95% confidence interval, p -value, and the incidence rate ratio and percent change of Equation 2.

4.2.5 Model comparison diagnostics

The choice of Negative Binomial over Poisson (Equation 3) is re-verified with the standard model-comparison toolkit rather than relying on the dispersion test alone: log-likelihood, AIC, BIC, Pearson χ^2/df , and a discrete goodness-of-fit (GOF) test. Because neither model’s cumulative distribution function is continuous, a plain continuous-KS diagnostic does not apply directly; GOF is instead assessed via randomized probability- integral-transform (PIT) values — for each observation, drawing $u \sim \text{Uniform}(F(y-1), F(y))$ under the fitted model’s CDF — pooled and Kolmogorov-Smirnov-tested against $\text{Uniform}(0, 1)$. This full comparison (Table 8) is run on **both** the full window and the post-ramp window, so the model choice is checked again on the smaller, more homogeneous 2025–2026 subset rather than carried over by default.

4.2.6 Potential software-release effect

The sharp reduction in PUDO complaint rates after the 2024 launch period temporally coincides with a major Waymo Driver software update. In a subsequent NHTSA recall filing, Waymo identified a new 5th-generation Automated Driving System (ADS) driverless software release dated November 7, 2024. Waymo reported that software improvements were deployed across the fleet through December 26, 2024, and that the release significantly reduced the likelihood of collisions with chains, gates, and similar roadway barriers [3]. NHTSA’s recall acknowledgment independently identifies the November 7 release and confirms that all affected vehicles had received the updated ADS software by December 26, 2024 [4].

Thus, 2025 Q1 is the first full analysis quarter following completion of this software deployment. The timing provides a plausible engineering hypothesis for a structural change in observed complaint rates; however, the documentation does not establish that the release specifically changed PUDO behavior or caused the observed decline. Treat this as a temporal association and candidate intervention, not causal evidence.

4.2.7 Launch-period sensitivity

2024 Q3–Q4 read as a deployment/launch ramp, with substantially higher exposure-adjusted rates than 2025 onward (Figure 8). Rather than simply discarding those quarters, Equation 4 is re-estimated — same model form, same trip-exposure offset — on two pre-registered windows to test whether they drive the full-period result:

- **A. Full period** — 2024 Q3 – 2026 Q2 ($n = 8$; the primary model above)
- **B. Post-ramp** — 2025 Q1 – 2026 Q2 ($n = 6$; excludes both 2024 quarters)

Comparing β_1 , its confidence interval, and its p -value across the two windows indicates whether the apparent decline is concentrated in the launch period or continues afterward. Given

$n = 8$, a simple two-window comparison is used rather than a piecewise or launch-indicator model, to avoid introducing more parameters than the data can support.

4.2.8 Exact (Garwood) Poisson confidence interval

Uncertainty around each **observed** quarter's count is reported with the exact Poisson interval rather than the normal/Wald approximation, so it stays valid at small or zero counts:

$$\text{CI}_{1-\alpha}(k) = \left[\frac{1}{2}\chi_{\alpha/2}^2(2k), \frac{1}{2}\chi_{1-\alpha/2}^2(2k+2) \right] / E \times 10^5 \quad (5)$$

where $\chi_p^2(\nu)$ is the p -quantile of a chi-square distribution with ν degrees of freedom, k the observed complaint count, and E the exposure (trips) for that quarter. This interval is **descriptive**: it summarizes uncertainty around one observed quarter's count/rate. It is distinct from — and not a substitute for — Equation 4's *inferential* uncertainty about the trend across quarters; overlapping exact Poisson intervals between quarters do not, by themselves, establish that no trend exists.

4.2.9 Distribution-free trend check (robustness)

As a **robustness check** that assumes no particular count distribution — not a replacement for the primary Negative-Binomial fit — the Mann-Kendall trend test [5], [6] computes Kendall's τ between the quarter index and the per-quarter rate $r_1, \dots, r_n$ (chronological order):

$$\tau = \frac{2}{n(n-1)} \sum_{i < j} \text{sign}(r_j - r_i) \quad (6)$$

$\tau < 0$ agrees with a declining trend; its p -value is reported alongside, separately from the NB model's.

4.2.10 Bayesian Negative-Binomial refit

Because only eight quarterly observations are available, Equation 4 is additionally estimated in a Bayesian framework to more fully characterize parameter *and* dispersion uncertainty — the frequentist fit instead estimates α by maximum likelihood and then treats it as *known*, relying on a large-sample approximation that a sample this small may not support. The same likelihood and exposure structure as Equation 4 is refit as a full posterior (PyMC [7], NUTS sampler) with weakly informative priors:

$$\begin{aligned} \beta_0 &\sim \mathcal{N}(\log(\bar{r}), 2^2) \\ \beta_1 &\sim \mathcal{N}(0, 1^2) \\ \alpha &\sim \text{Exponential}(1) \\ \mu_t &= E_t \exp(\beta_0 + \beta_1 t) \\ Y_t \mid \mu_t, \alpha &\sim \text{NB}(\mu_t, \alpha), \quad \text{Var}(Y_t) = \mu_t + \mu_t^2/\alpha \end{aligned} \quad (7)$$

where $\bar{r}$ is the pooled baseline rate and $\beta_1 \sim \mathcal{N}(0, 1)$ deliberately expresses no prior belief about direction. Note Equation 7’s variance term is the **inverse** sense of Equation 4’s α (larger α here means *closer* to Poisson) — the two are not numerically comparable. The model is sampled with 4 chains $\times$ 2000 draws, checked for convergence ($\hat{R}$, effective sample size, divergence count) and for sensitivity to the prior width on β_1 . The quantity of interest is the posterior probability the rate is actually declining, estimated directly from the fraction of posterior draws below zero rather than from a p -value:

$$P(\text{declining}) = P(\beta_1 < 0 \mid y_{1:n}) \quad (8)$$

This Bayesian refit is reported as a robustness / small-sample analysis alongside the frequentist estimate, not as a claim that it fits the data better: no model-comparison criterion (e.g. WAIC/LOO) or posterior-predictive check is computed here, so no such claim is made.

4.2.11 Latest-quarter contrast

The main question this update adds is whether the **most recent** quarter differs from the rest of the post-ramp window, controlling for exposure. Complaints carry no time resolution finer than the reporting quarter (confirmed directly against the underlying data, not assumed), so the latest quarter is exactly one observation — comparing one observation against a handful of quarterly aggregates with a rank test (Mann-Whitney U) or a t -test is not a valid two-sample test. Two regression-based contrasts are used instead, both fit on the post-ramp window:

1. A **regime-indicator Negative-Binomial regression**, $Y_t \sim \text{NB}(\mu_t, \alpha)$ with $\log \mu_t = \log E_t + \beta_0 + \beta_{\text{latest}} \mathbb{1}[t = \text{latest}]$, testing $H_0 : \beta_{\text{latest}} = 0$ via a Wald test, reported with its incidence rate ratio (effect size), 95% CI, and an approximate post-hoc power at the observed effect size (illustrative only, given $n = 6$).
2. A **leave-latest-out predictive check**: Equation 4 refit on the post-ramp quarters *preceding* the latest one, producing a predictive rate and 95% interval (delta method) for what the latest quarter should look like under the established recent pattern; the actually-observed rate is checked against that interval.

The same regime-indicator model is additionally fit in the Bayesian framework of Equation 7 (weakly informative priors, PyMC/NUTS), reporting the posterior probability the latest quarter’s rate is higher or lower than the rest of the post-ramp window, for comparison against the frequentist contrast.

4.2.12 Risk-matrix probability basis

To place the PUDO hazard on a probability $\times$ impact risk matrix, three ways of expressing “how likely is a PUDO event” are compared, holding the impact reading fixed. Let $k = \sum_t y_t$ be total complaints and $N = \sum_t E_t$ total trips over the window.

Per-period — the probability of at least one complaint in a reporting quarter, treating the mean quarterly count $\bar{\lambda} = k/n$ as a Poisson rate:

$$p_{\text{per-period}} = 1 - e^{-\bar{\lambda}} \quad (9)$$

Per-trip — the raw empirical per-trip probability:

$$p_{\text{per-trip}} = \frac{k}{N} \quad (10)$$

Bayesian — a Beta-Binomial posterior on the per-trip rate (Jeffreys-style Beta(0.5, 0.5) prior), converted to a predictive probability of at least one complaint in a future quarter of size E_{next} :

$$\begin{aligned} \text{rate} \mid k, N &\sim \text{Beta}(0.5 + k, 0.5 + N - k) \\ p_{\text{bayesian}} &= 1 - e^{-\widehat{\text{rate}} \cdot E_{\text{next}}} \end{aligned} \quad (11)$$

using the posterior mean $\widehat{\text{rate}}$ and the most recent quarter’s trip volume as E_{next} .

5 Evaluation and Results

Per-quarter descriptive statistics for the complaint counts, exposure, and rate (Table 2) show why raw counts cannot be compared directly: exposure grows nearly 6× across the window.

Table 2: Per-quarter descriptive statistics for PUDO complaints, trips, and the PUDO-per-100k-trips rate, 2024 Q3 - 2026 Q2 (n = 8).

variable	mean	std	min	median	max	range	CV
PUDO complaints	52.125	17.57	27.0	46.5	79.0	52.0	0.337
trips (M)	2.651	1.216	0.752	2.455	4.22	3.468	0.459
rate /100k trips	2.317	1.324	1.485	1.819	5.453	3.968	0.572

That growth is not just quarter-to-quarter: Figure 1 shows the real within-quarter spread in monthly trip volume (median and interquartile range built from the underlying monthly observations, not a synthetic range from a single quarterly total). PUDO complaints have no equivalent view — every complaint record is dated only to its reporting quarter, not to a month within it — so only the exposure side of this analysis can be shown at this resolution.

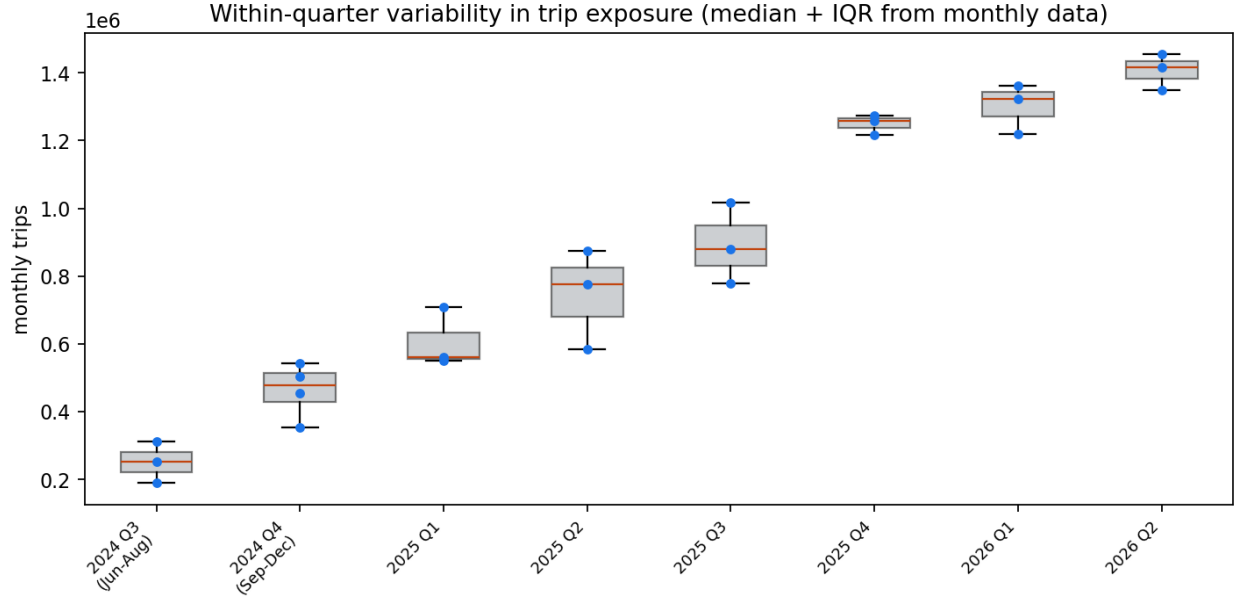


Figure 1: Within-quarter variability in trip exposure: monthly trips, median and IQR per analysis quarter.

The index-of-dispersion test (Equation 3) on these counts gives $\text{VMR} \approx 5.9$ ($D \approx 41.5$ against χ^2_7 , $p < 0.001$), rejecting Poisson equidispersion — the reason every trend fit below uses the Negative-Binomial model of Equation 4 rather than Equation 1 directly. The per-quarter rates in Table 3 and throughout use the exact interval of Equation 5, not a normal approximation.

Pooling across all eight quarters gives the headline exposure-adjusted rate (Table 3): Over the study window Waymo averaged **1.97** PUDO complaints per 100,000 trips (exact Poisson 95% CI 1.78-2.16), from 417 complaints over 21.2M trips.

Table 3: Pooled PUDO complaint rate over the study window (417 complaints, 2024 Q3 - 2026 Q2).

complaints	trips (M)	rate per 100k trips	95% CI
417	21.2	1.966	[1.782, 2.164]

PUDO complaints are also worth reading in context of everything else riders report. Figure 2 stacks the PUDO count against every other reported complaint category, per quarter:

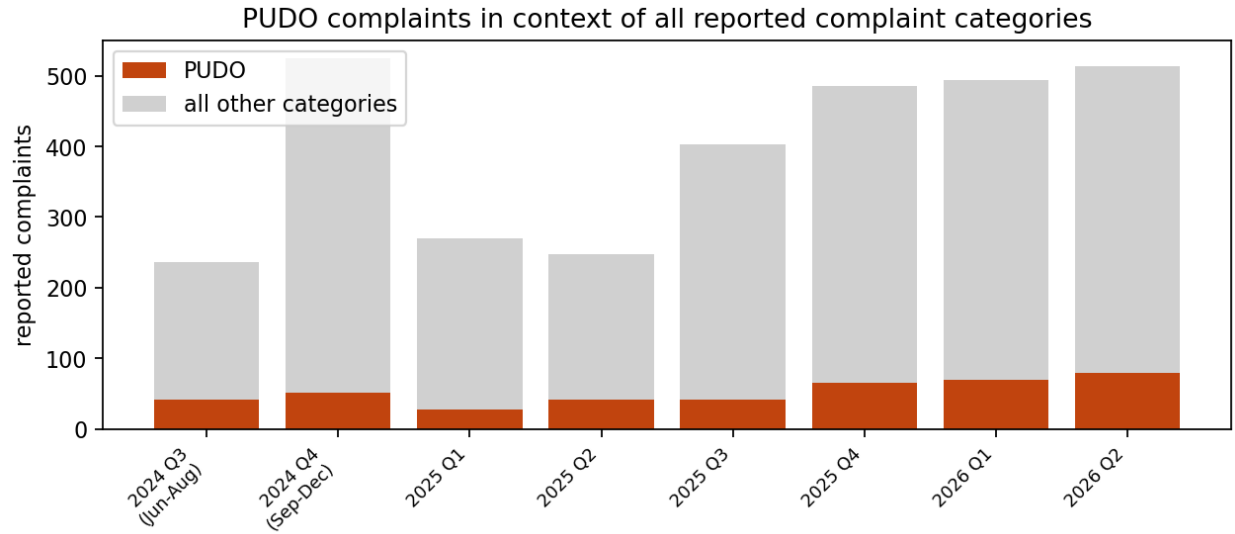


Figure 2: PUDO complaints against all other reported complaint categories, per quarter.

PUDO complaints average **13%** of all reported complaint categories over the window (range 10%-17%), making PUDO the largest location/behaviour complaint category (Safety and ‘Other’ are larger overall categories, but are not location-specific).

PUDO complaints scale with total complaint volume — the PUDO share stays in a fairly narrow band rather than drifting — so PUDO is not a shrinking or growing slice of the overall complaint mix; it is consistently the largest category tied to a specific vehicle maneuver rather than a general safety or “other” catch-all.

5.0.1 Distributional diagnostics: full vs. post-ramp

The same per-quarter statistics are also computed for the post-ramp window (2025 Q1 – 2026 Q2, $n = 6$) used throughout the Launch-period sensitivity and Latest-quarter results below:

Table 4: Per-quarter descriptive statistics for PUDO complaints, trips, and the PUDO-per-100k-trips rate, post-ramp window (2025 Q1 - 2026 Q2, $n = 6$).

variable	mean	std	min	median	max	range	CV
PUDO complaints	54.167	20.054	27.0	54.0	79.0	52.0	0.37
trips (M)	3.101	0.99	1.818	3.212	4.22	2.402	0.319
rate /100k trips	1.722	0.161	1.485	1.764	1.879	0.394	0.094

Figure 3 and Figure 4 show the quarterly PUDO rate’s distribution in both windows; Table 5 reports the Shapiro-Wilk normality test for each (H_0 : the rate values are normally distributed): Shapiro-Wilk test for normality on the quarterly PUDO rate: Full: $W=0.639$, $p=0.001$ (REJECT normality); Post-ramp: $W=0.883$, $p=0.282$ (fail to reject normality).

Table 5: Shapiro-Wilk normality test on the quarterly PUDO rate, full vs. post-ramp window.

window	n	W	p-value	verdict
Full (n=8, 2024Q3-2026Q2)	8	0.639	0.0005	REJECT normality
Post-ramp (n=6, 2025Q1-2026Q2)	6	0.883	0.2818	fail to reject normality

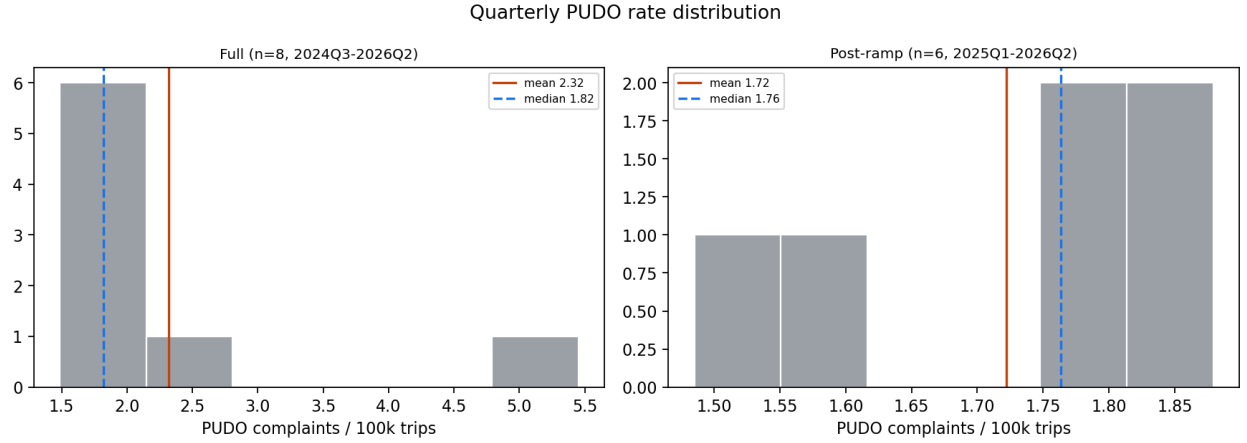


Figure 3: Quarterly PUDO rate distribution, full vs. post-ramp window.

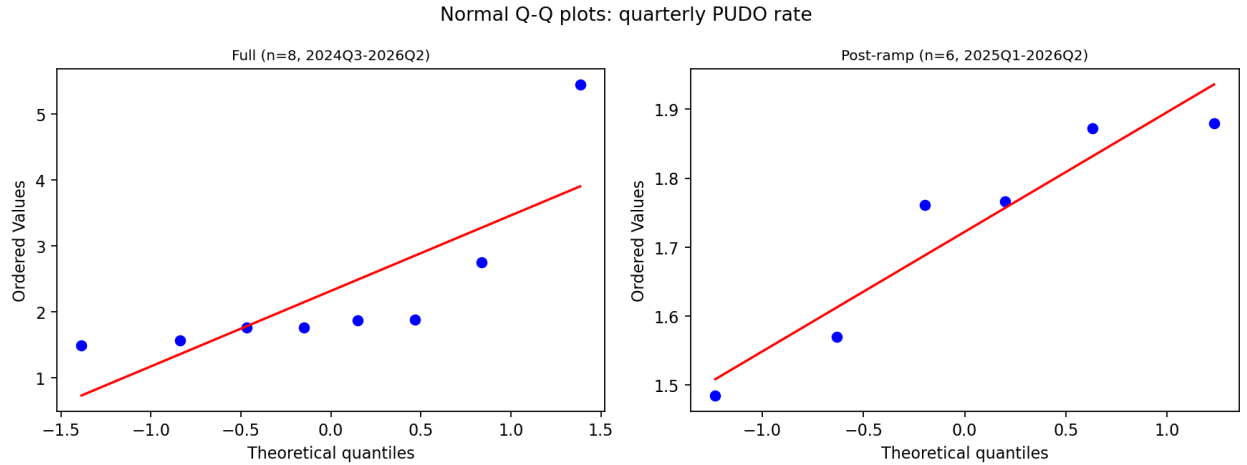


Figure 4: Normal Q-Q plots: quarterly PUDO rate, full vs. post-ramp window.

With $n = 6-8$ and, in the full window, a single extreme launch-period outlier (2024 Q3), this is exactly why classic parametric mean-comparison tests are not used anywhere in this paper for the complaint rate — every trend and contrast in this paper instead models the rate through the count-model (Poisson/Negative-Binomial) framework, which treats the data as counts over exposure rather than assuming the rate itself is Gaussian.

Figure 5 compares the empirical CDFs of the 2024 launch-quarter rates against the 2025–2026 rates directly; with $n = 2$ against $n = 6$ this two-sample KS comparison is descriptive only, not a well-powered test.

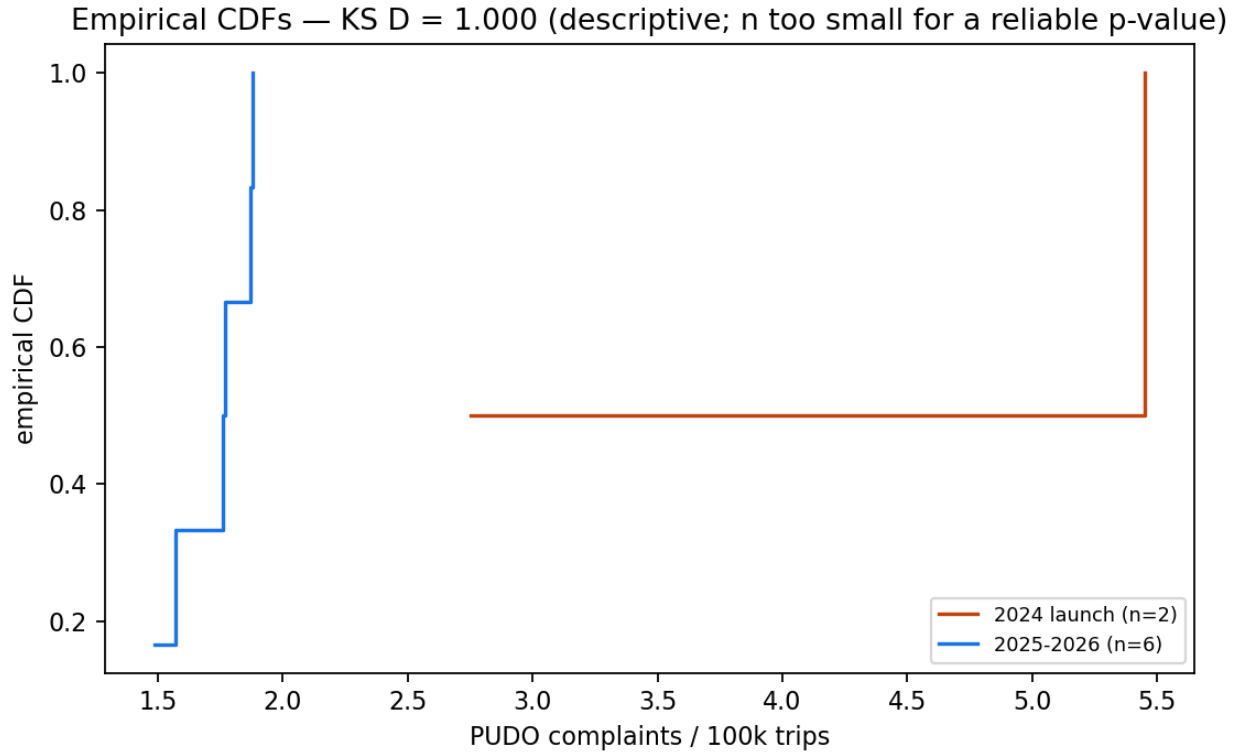


Figure 5: Empirical CDFs: 2024 launch-quarter PUDO rate vs. 2025–2026 PUDO rate (descriptive; n too small for a reliable p -value).

A better-powered two-sample KS comparison is available for monthly trip exposure, where 2025 and 2026 each contribute a real per-month sample ($n = 12$ and $n = 6$). Figure 6 and Figure 7 show the Q-Q plots and empirical CDFs used to check the normality assumption behind a companion Welch two-sample t -test comparing monthly 2025 and 2026 trip volumes directly (reported in the companion notebook) — a check on trip *volume* growth, distinct from the exposure-adjusted complaint-rate contrast in “Is the latest quarter different?” below.

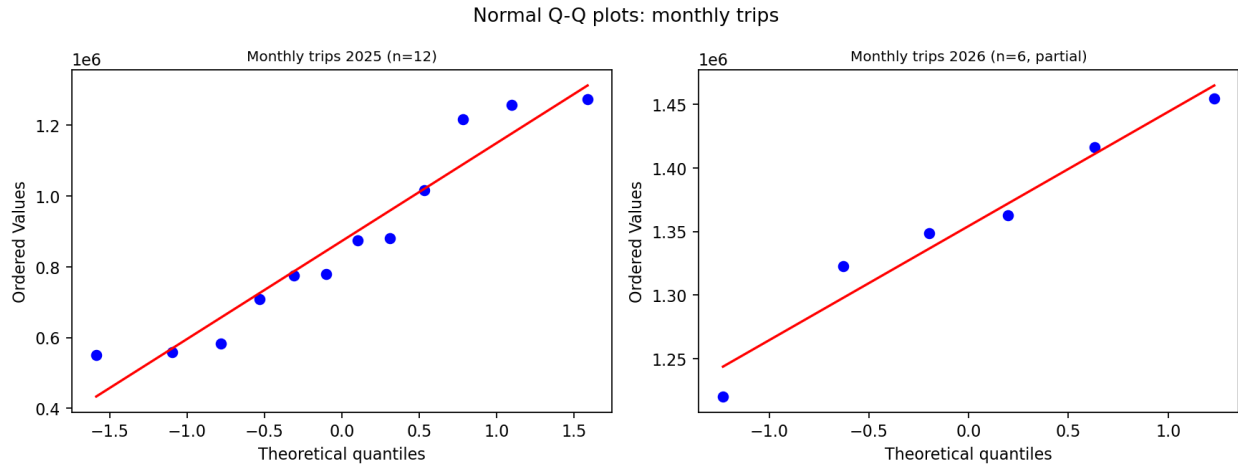


Figure 6: Normal Q-Q plots: monthly trips, 2025 vs. 2026.

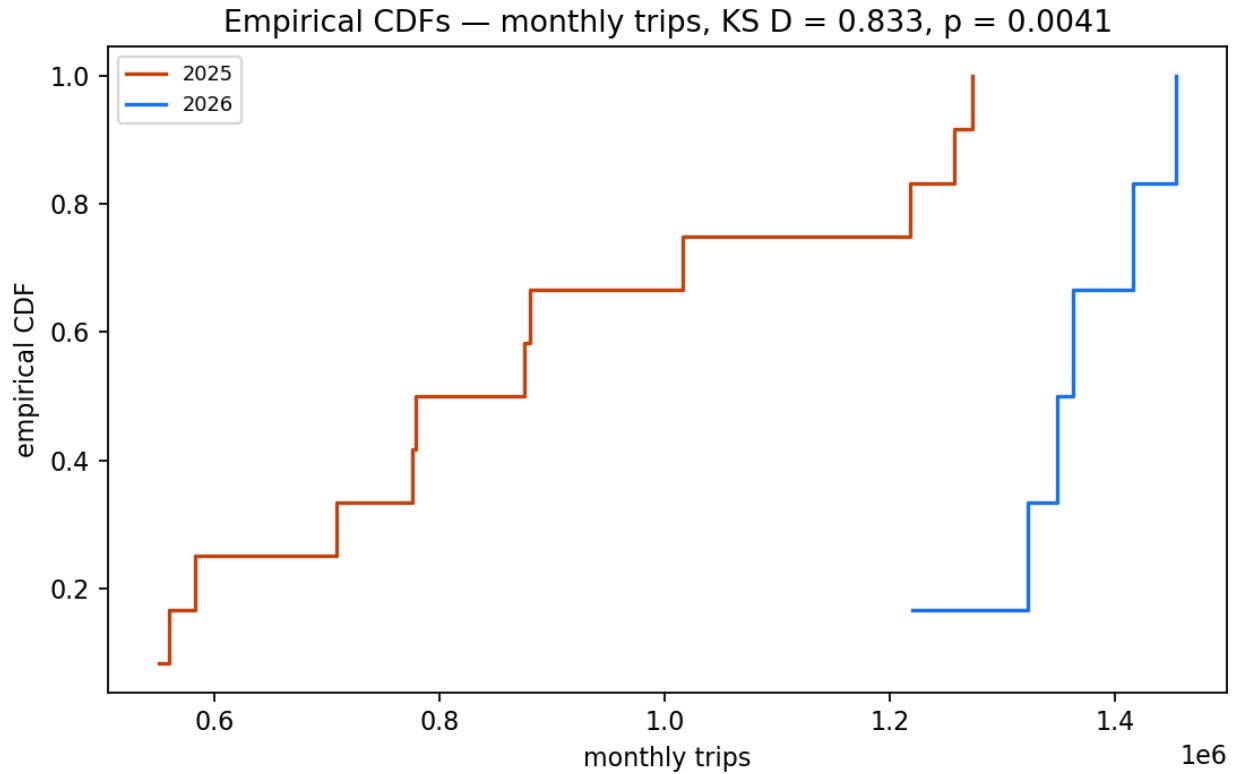


Figure 7: Empirical CDFs: monthly trips, 2025 vs. 2026.

5.0.2 Primary trend: Negative-Binomial regression

The primary trend model — Equation 4 fit on trips exposure — gives (Table 6): The primary trend model is Negative-Binomial regression (`pudo_complaints ~ t, offset = log(trips)`): $\text{beta}_t = -0.114$ (SE 0.042, 95% CI [-0.196, -0.032]), an incidence rate ratio

of 0.893 per quarter – a **-10.7%** change in the PUDO complaint rate per quarter (95% CI -17.8% to -3.1%, $p = 0.007$; total change -55% across the window).

Table 6: Primary Negative-Binomial trend model ($\text{pudo_complaints} \sim t$, $\text{offset} = \log(\text{trips})$): coefficient, standard error, confidence interval, incidence rate ratio, and percent change per quarter.

term	beta_t SE	95% CI (log)	IRR = exp(beta_t)	% change / qtr	95% CI (%) / qtr	p-value
t (quarter index)	-0.0419 0.1137	[-0.1958, -0.0315]	0.8926	-10.7%	[-17.8%, -3.1%]	0.0067

In plain language: each additional quarter multiplies the expected PUDO complaint rate per trip by the incidence rate ratio (IRR) in Table 6, holding exposure fixed. The Poisson diagnostic fit (Equation 1) is not reported here as an inferential result — it assumes equidispersion, which Equation 3 rejects.

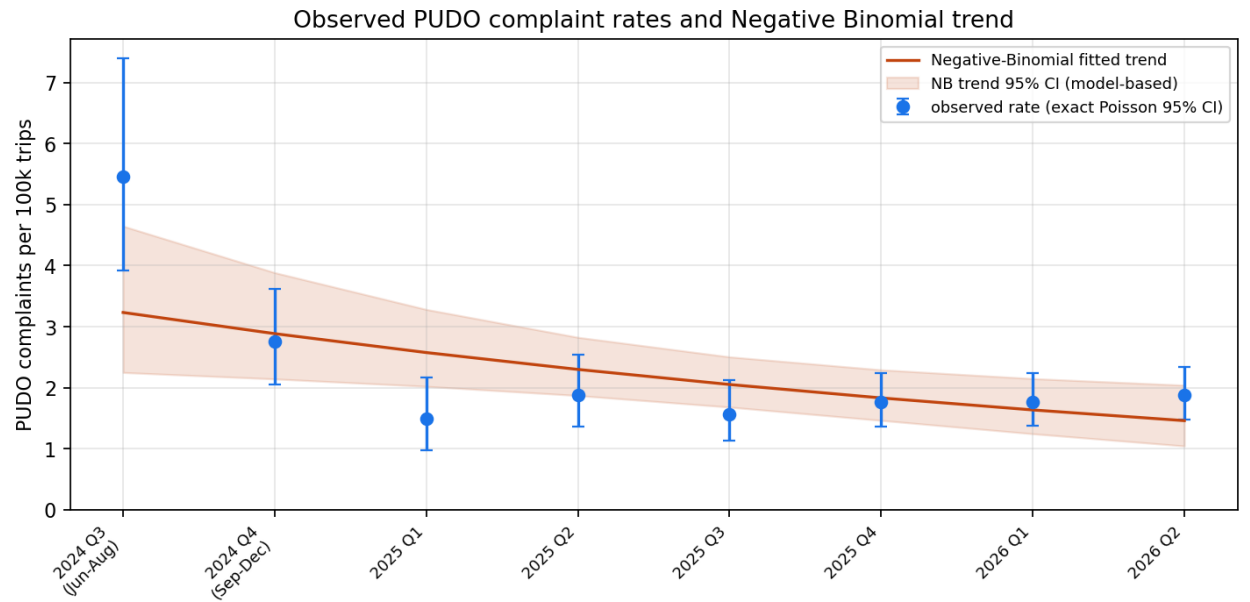


Figure 8: Observed PUDO complaint rate per 100k trips (exact Poisson 95% CI, a descriptive interval around each observed quarter) against the Negative- Binomial fitted trend and its own 95% model-based confidence band.

Figure 8 keeps the two kinds of uncertainty visually distinct: the blue error bars are Equation 5, describing each observed quarter’s count; the red band is the NB regression’s uncertainty about the *trend line* itself. The two 2024 quarters sit well above the rest of the series and above the fitted line’s lower band — a sharp drop from a launch-period high, followed by a rate that moves within a fairly narrow range for the remaining six quarters. The overlapping

bars from 2025 onward describe individual quarters and are not by themselves evidence for or against a trend; that question is addressed directly below.

5.0.3 Launch-period sensitivity

Restricting the identical NB model to the post-ramp window tests whether the full-period decline is concentrated in the 2024 launch quarters (Table 7): Restricting the primary Negative-Binomial model to the post-ramp window (2025 Q1 - 2026 Q2, $n = 6$) gives a slope of **+2.9%** per quarter (95% CI -3.8% to +10.2%, $p = 0.402$), versus **-10.7%** per quarter ($p = 0.007$) over the full study period. The apparent decline over the full study period appears to be driven substantially by the elevated rates observed during the 2024 launch period: once those two quarters are excluded, the remaining six quarters show a much weaker, no-longer-significant trend. Rates from 2025 onward appear comparatively stable. This is consistent with a launch/ramp-period effect rather than a continuing decline, though it does not establish causation.

Table 7: Negative-Binomial trend sensitivity: full study period vs. the post-ramp window excluding both 2024 launch-period quarters.

window	n	beta_t	SE	% / qtr	95% CI	p- value	total change
A. Full (2024Q3-2026Q2)	8	-	0.042	-	[-17.8%, -3.1%]	0.007	-55%
B. Post-ramp (2025Q1-2026Q2)	6	0.029	0.035	+2.9%	[-3.8%, +10.2%]	0.402	+16%

The post-ramp slope is close to zero and not statistically significant on its own, in contrast to the full-period estimate. The apparent decline over the full study period appears to be driven substantially by the elevated rates observed during the 2024 launch period, while rates from 2025 onward appear comparatively stable — consistent with a launch/ramp-period effect rather than a continuing decline, though this does not establish causation.

5.0.4 Model comparison: Poisson vs. Negative Binomial (full vs. post-ramp)

Section 1's dispersion-test-only justification for Negative Binomial (Equation 3) is checked again here with the fuller model-comparison toolkit described in Methods, on **both** windows (Table 8): Re-verifying (not assuming) the Poisson-vs-Negative-Binomial choice with the full comparison toolkit (log-likelihood, AIC, BIC, Pearson dispersion, discrete goodness-of-fit): Full ($n=8$): lowest AIC/BIC $\rightarrow$ Neg. binomial; Post-ramp ($n=6$): lowest AIC/BIC $\rightarrow$ Poisson.

Table 8: Poisson vs. Negative-Binomial model comparison (log-likelihood, AIC, BIC, Pearson dispersion, discrete GOF KS test), full vs. post-ramp window.

window	model	n	k	llf	AIC	BIC	pear- son_chi2_df	KS_D	KS_p
Full (n=8)	Poisson	8	2	- 37.12	78.23	78.39	5.15	0.2378	0.6723
Full (n=8)	Neg. binomial	8	3	- 32.92	71.85	72.09	5.03	0.1395	0.9908
Post-ramp (n=6)	Poisson	6	2	- 17.88	39.76	39.34	0.29	0.2901	0.5979
Post-ramp (n=6)	Neg. binomial	6	3	- 17.88	41.76	41.14	0.29	0.2901	0.5978

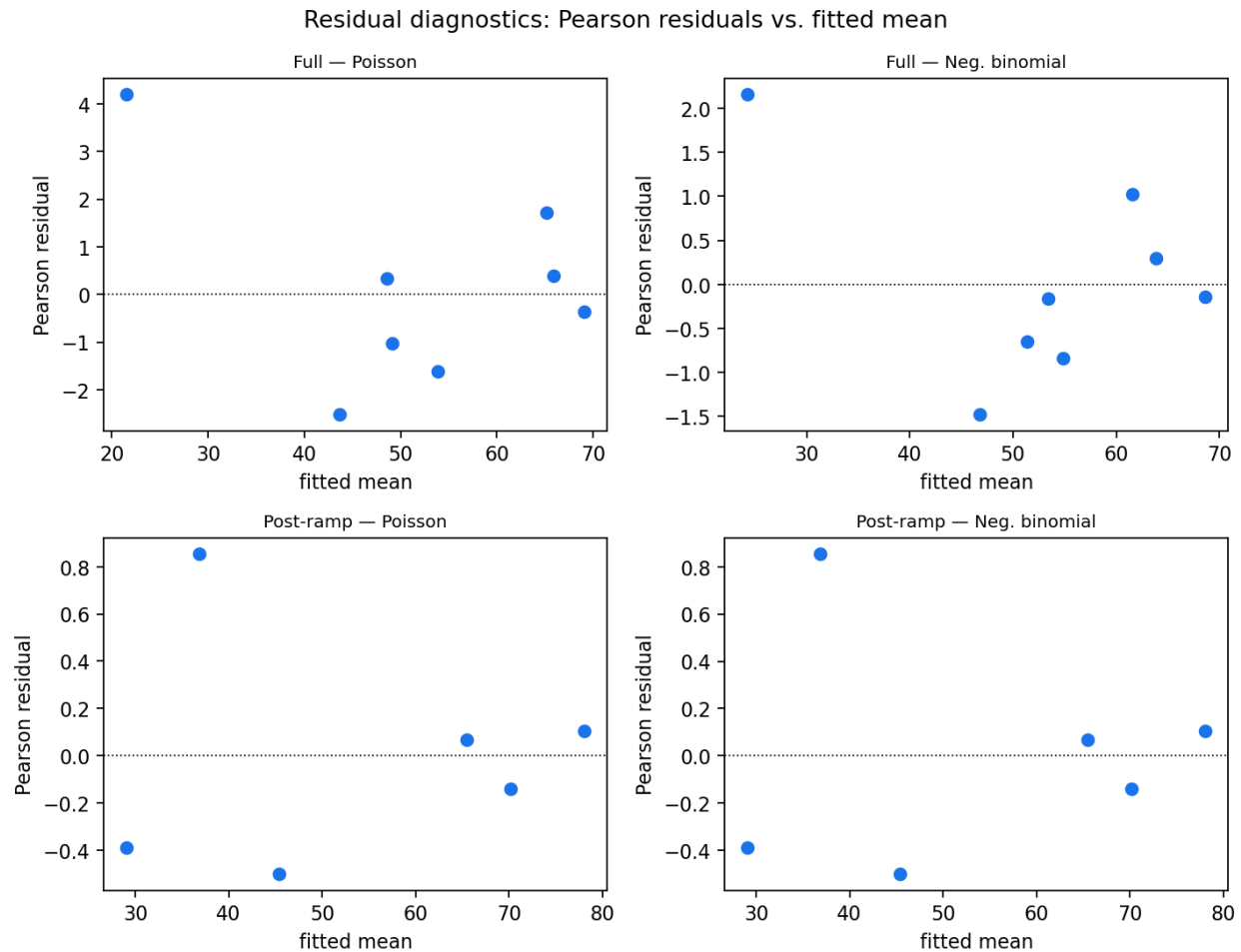


Figure 9: Residual diagnostics: Pearson residuals vs. fitted mean, for each model/window combination.

The result is not assumed to carry over unchanged: the post-ramp window is smaller and

— per the Pearson χ^2/df column — far less overdispersed than the full window, since it excludes the volatile 2024 launch quarters. When a window shows little real overdispersion, Negative Binomial’s extra dispersion parameter earns its AIC/BIC complexity penalty less easily than in the full window; Table 8 reports whichever model actually wins in each case, rather than carrying the full-window conclusion over by assumption. Figure 9 shows no systematic pattern in the Pearson residuals for either model in either window. Either way, the discrete GOF test finds no strong evidence against either model’s fit in either window at this sample size.

5.0.5 Robustness checks

Refitting Equation 4 under the VMT-total cross-check gives the same direction as the primary trips-based fit, and Kendall’s τ (Equation 6) — a distribution-free check, not an alternative primary model — agrees in sign under both denominators (Table 9): The primary Negative-Binomial trend is negative under all 2 exposure denominators tested (per-quarter change -13% to -11%; NB p 0.001-0.007). Mann-Kendall’s tau, a distribution-free robustness check, is negative under both denominators as well, though only marginally significant (p 0.40-0.55) given the non-monotone, fell-then-flat shape of the series.

Table 9: Primary Negative-Binomial trend fit and Mann-Kendall robustness check under every exposure denominator.

denomina- tor	model	beta_t	SE	%/qtr	95% CI	p (NB)	total	MK tau	p (MK)
Trips	Neg.	-	0.042	-	[-18%,	0.007	-	-0.21	0.548
	binomial	0.114		10.7%	-3%]		55%		
VMT total	Neg.	-	0.042	-	[-20%,	0.001	-	-0.29	0.399
	binomial	0.135		12.7%	-5%]		61%		

Mann-Kendall’s weaker significance reflects the non-monotone, fell-then-flat shape of the series (limited power for a monotone-rank test on $n = 8$), not evidence that the trend is absent.

5.0.6 Bayesian robustness check

The Bayesian refit (Equation 7, Table 10) is reported alongside the primary frequentist estimate as a small-sample robustness check — not a claim that it fits the data better — and reports Equation 8 directly rather than a p -value: A Bayesian Negative-Binomial fit (weakly informative priors, PyMC/NUTS, 4 chains x 2000 draws, 0 divergences, max R-hat 1.00) gives a posterior median slope of **-10.9%** per quarter (95% credible interval [-25.2%, +7.6%]) and **P(the rate is declining) = 90%** — consistent with, but appropriately less certain than, the frequentist $p = 0.007$.

Table 10: Bayesian Negative-Binomial trend model posterior summary (PyMC/NUTS, 4 chains x 2000 draws, 0 divergences).

parameter	mean	sd	hdi_2.5%	hdi_97.5%	r_hat	ess_bulk
b0	-10.282	0.394	-11.059	-9.529	1.0	2731.0
b1	-0.115	0.091	-0.299	0.058	1.0	2741.0
alpha	3.131	1.502	0.649	6.036	1.0	2921.0

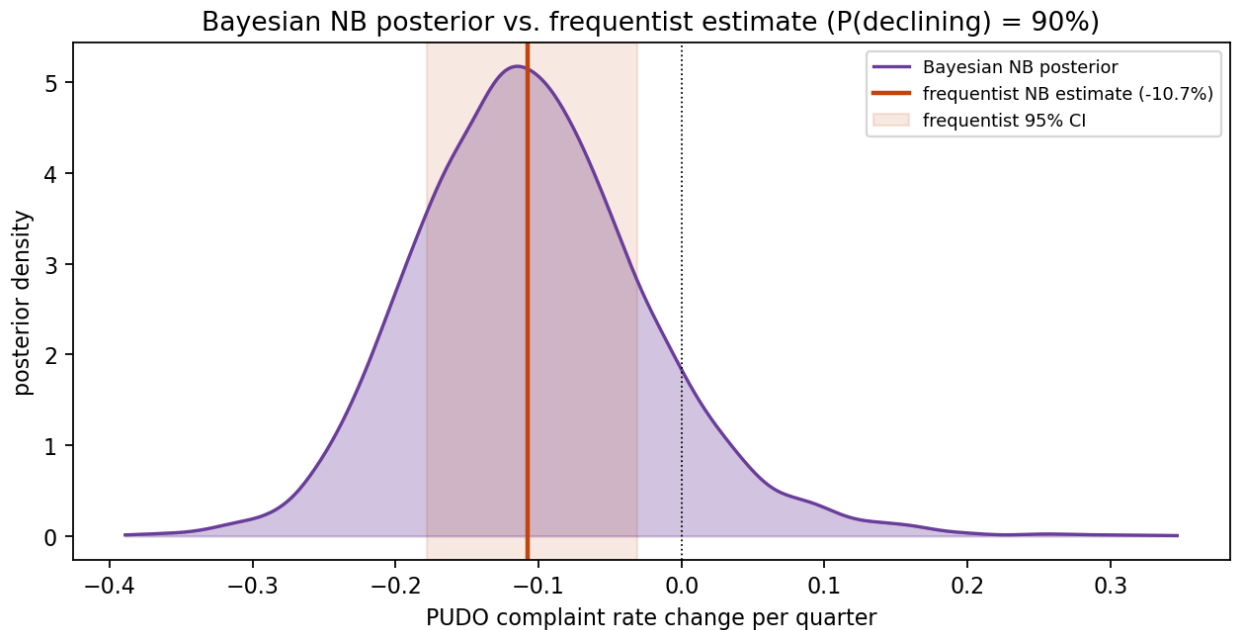


Figure 10: Bayesian Negative-Binomial posterior on the per-quarter rate change, vs. the frequentist point estimate and 95% CI.

The posterior barely shifts across a $4\times$ range of prior width on the slope (Table 11) — the conclusion is driven by the data, not the choice of prior:

Table 11: Bayesian NB posterior sensitivity to the prior width on the trend slope.

prior sd on slope	posterior median %/qtr	P(declining)
0.5	-11.0%	90.0%
1.0	-10.8%	90.6%
2.0	-11.1%	90.6%

5.0.7 Is the latest quarter different?

This update’s main question: controlling for trip exposure, does the most recent quarter look different from the rest of the post-ramp window? As established in Methods, complaints

have no time resolution finer than the reporting quarter, so the latest quarter is exactly one observation and a rank test (Mann-Whitney U) or t -test comparing it against a handful of quarterly aggregates would not be a valid two-sample test — the two regression-based contrasts of the Latest-quarter contrast methods section are used instead (Table 12): The regime-indicator contrast (Poisson GLM, since NB’s alpha converged to the Poisson boundary) finds the 2026Q2 PUDO rate does not differ significantly from the rest of the post-ramp window (IRR 1.095, 95% CI [0.849, 1.410], $p = 0.485$). The leave-latest-out predictive check finds the observed rate (1.872/100k trips) falls **within** the 95% predictive interval built from the preceding quarters ([1.391, 2.376]) – no evidence of departure from the recent stable pattern.

Table 12: Is the latest available quarter different from the preceding post-ramp quarters? Regime-indicator NB regression and leave-latest-out predictive check.

latest quar- ter	beta_ est	lat- SE	z	p- value	95% CI IRR(IRR)	ap- prox. power	predicted rate (preceding- only fit)	pre- dicted 95% CI	ob- served rate	within predictive interval
2026Q2	0.0904	0.1293	0.699	0.484	1.095 [0.849, 1.410]	0.1	1.818	[1.391, 2.376]	1.872	True

The regime-indicator contrast’s effect size and confidence interval, and the leave-latest-out predictive check’s interval, both come from a six-quarter window — enough to rule out only a large shift, not a small one (see Limitations). The Bayesian regime-indicator fit (Table 13) reaches the same qualitative reading with an explicit posterior probability rather than a p -value: A Bayesian regime-indicator NB fit on the post-ramp window gives a posterior median IRR of **1.124** for 2026Q2 vs. the rest (95% credible interval [0.405, 3.842]), with $P(\text{higher}) = 59\%$ and $P(\text{lower}) = 41\%$ – agreeing with the frequentist regime-indicator contrast ($p = 0.485$).

Table 13: Bayesian Negative-Binomial regime check for the latest quarter within the post-ramp window.

parameter	mean	sd	hdi_2.5%	hdi_97.5%	r_hat	ess_bulk
b0	-10.936	0.297	-11.49	-10.314	1.0	4869.0
b_latest	0.141	0.571	-0.936	1.314	1.0	5531.0
alpha	3.213	1.7	0.59	6.639	1.0	4516.0

5.0.8 Risk curve & risk matrix

Where the trend result above answers *how the frequency is changing*, this subsection places PUDO risk on a probability $\times$ impact matrix — the risk-assessment framework this paper’s core question ultimately serves. The consequence side of that placement is read off a risk

curve: how the event rate falls as severity rises (Figure 11). Nothing severe or fatal was observed at the PUDO level over the window.

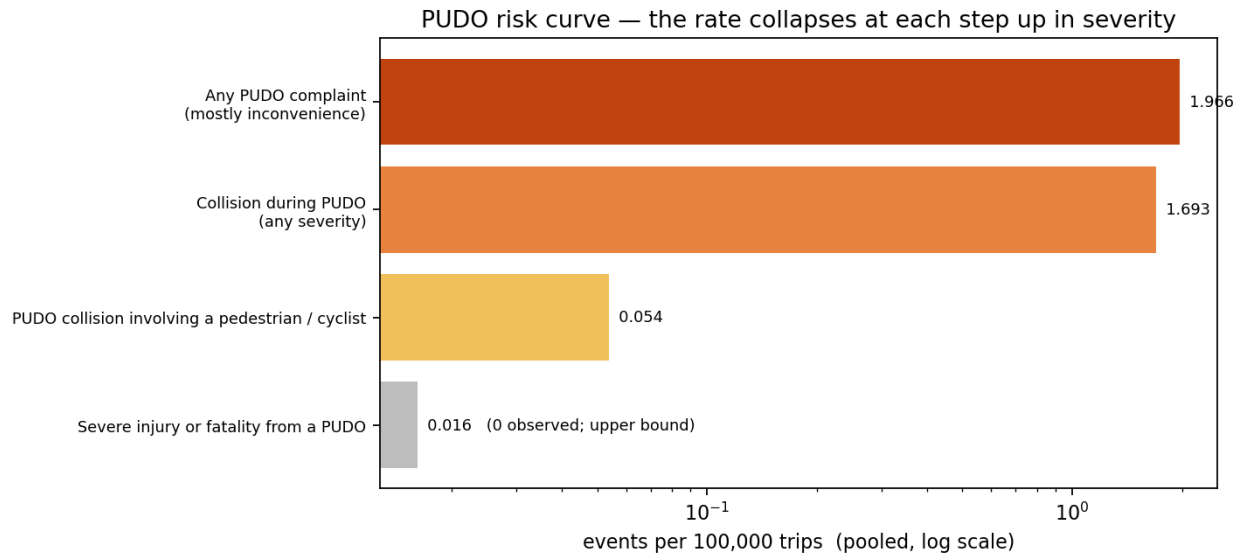


Figure 11: PUDO risk curve: event rate per 100,000 trips at each step up in consequence severity (log scale). The unfilled bar is a rule-of-three upper bound on a rate with zero observed events, not a measurement.

Combining that impact reading with a probability of a PUDO complaint occurring in a given reporting quarter — Table 14 compares the three formulas of Equation 9, Equation 10, and Equation 11 — places the hazard on the matrix as follows: Probability: **Almost certain** (posterior rate 1.97/100k trips (95% CrI 1.78-2.16); predictive $P(>=1 \text{ next quarter}) = 1.0000$). Impact: **Minor** (the worst consequence tier clearing 0.1/100k trips in the risk curve above). Combined risk level: **Medium**.

Table 14: PUDO risk-matrix placement under three probability bases (impact held fixed).

prob basis	probability	impact	risk level
per_period	Almost certain	Minor	Medium
per_trip	Moderate	Minor	Low
bayesian	Almost certain	Minor	Medium

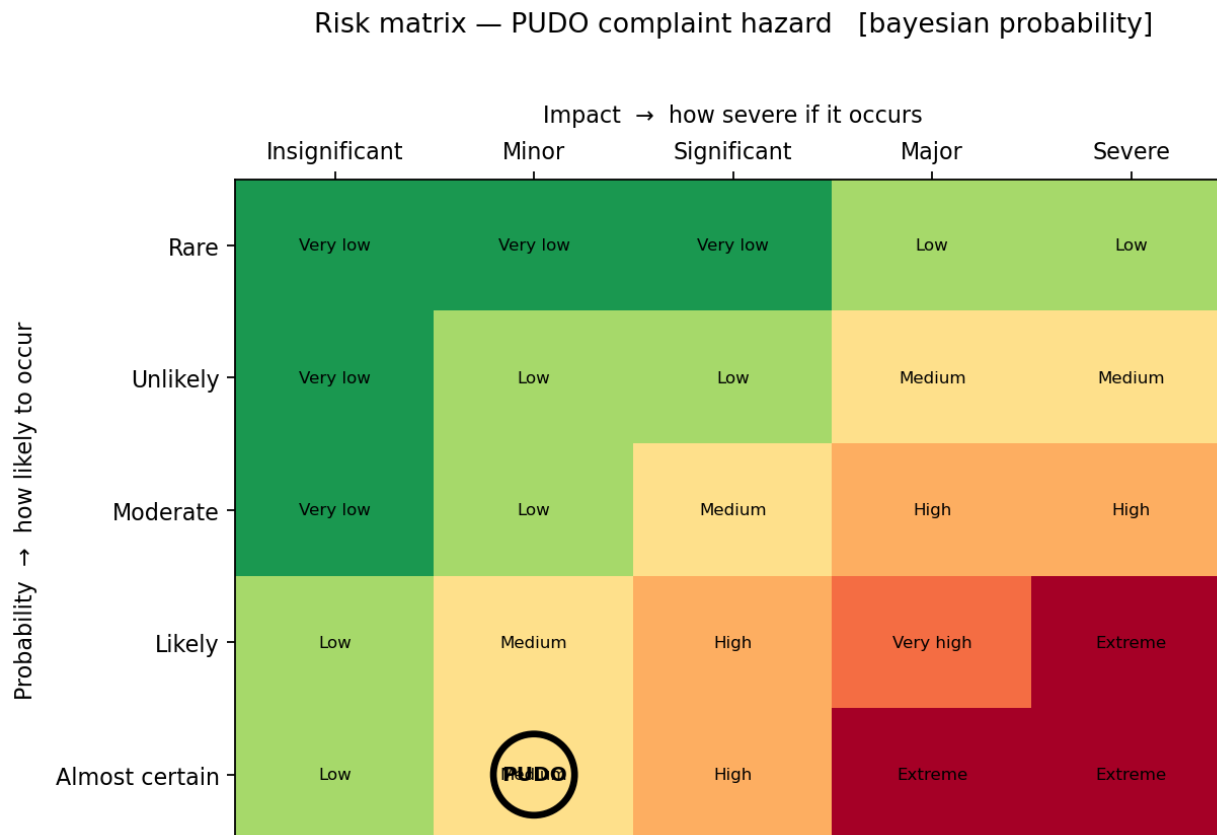


Figure 12: PUDO complaint hazard on a probability $\times$ impact risk matrix, placed from the data under the `per_period` probability basis.

5.0.9 Spatial PUDO risk

`AV_Incidents_Location` gives PUDO collision counts aggregated to the Census-tract level, per analysis quarter — the finest location this dataset ever resolves to (individual complaint/collision coordinates and `PUDOTravelLane` remain redacted, as throughout this paper). All 8 of 8 analysis quarters have usable location data, spanning two disjoint California service areas — the LA Basin and the SF Bay Area/Peninsula corridor — mapped as separate panels below rather than one map with empty Central Valley between them. Tract boundaries are U.S. Census Bureau TIGER/Line shapefiles [8] — both the 2020- and 2010-vintage boundaries are used, since CPUC/Waymo’s tract identifiers mix tracts from both eras — rendered on an OpenStreetMap basemap [9].

Table 15: Geographic coverage of AV_Incidents_Location by analysis quarter: tracts reporting, tracts with at least one PUDO collision, and pooled counts (all 8/8 analysis quarters have usable data).

period	tracts reporting	tracts w/ PUDO collision	PUDO collisions	all collisions
2024 Q3 (Jun-Aug)	1351	9	11.0	55.0
2024 Q4 (Sep-Dec)	1351	25	30.0	47.0
2025 Q1	1496	29	34.0	133.0
2025 Q2	1496	46	54.0	165.0
2025 Q3	1796	38	46.0	168.0
2025 Q4	1796	40	50.0	266.0
2026 Q1	1796	60	69.0	301.0
2026 Q2	1796	55	65.0	297.0

The two maps below are interactive in the HTML edition of this paper — pan, zoom, and hover any tract for its GEOID, PUDO collision count, total collision count, and which analysis quarters it reported in:

PUDO collisions by Census tract — LA Basin (2024 Q3 – 2026 Q2)

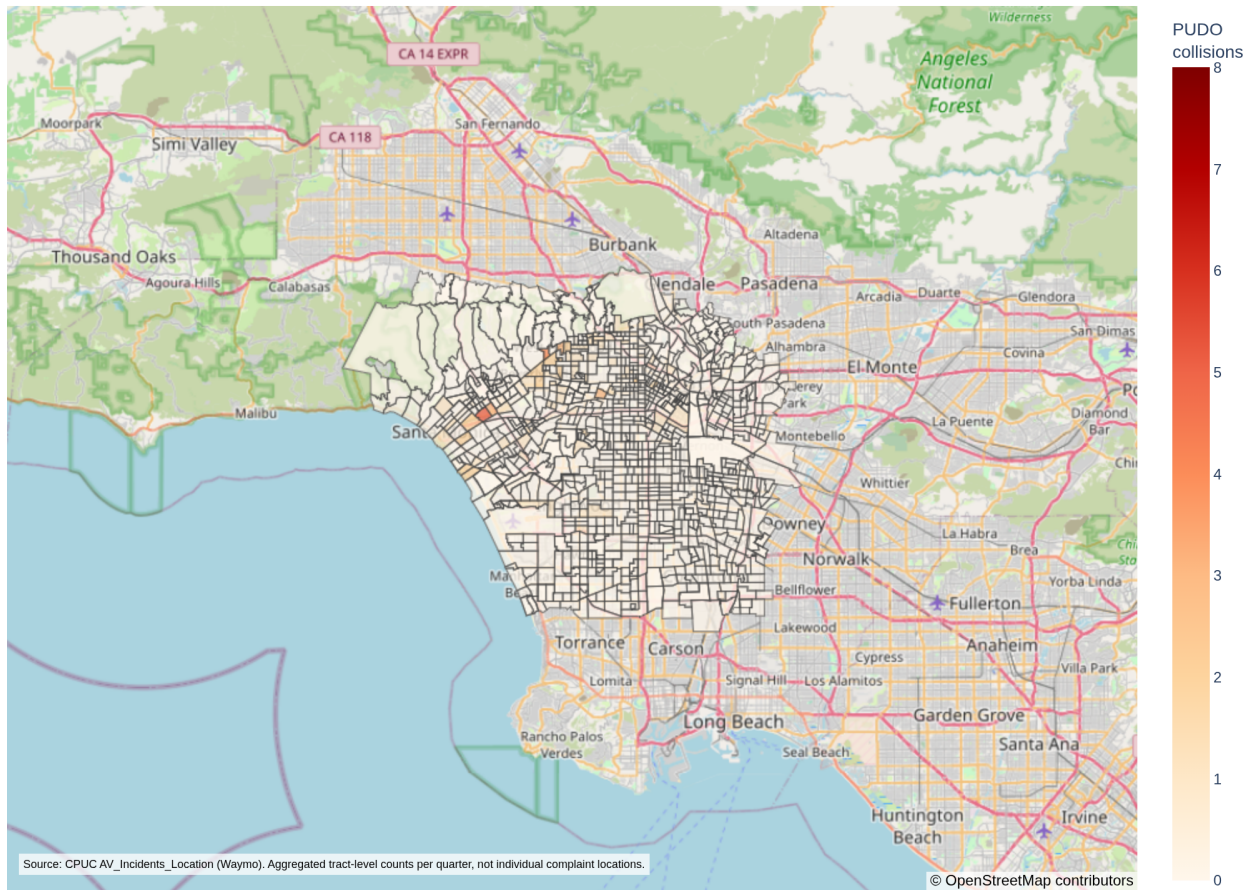


Figure 13: PUDO collisions by Census tract, LA Basin (2024 Q3 – 2026 Q2). The HTML edition of this paper has an interactive, pannable/zoomable version of this map.

PUDO collisions by Census tract — SF Bay Area (2024 Q3 – 2026 Q2)

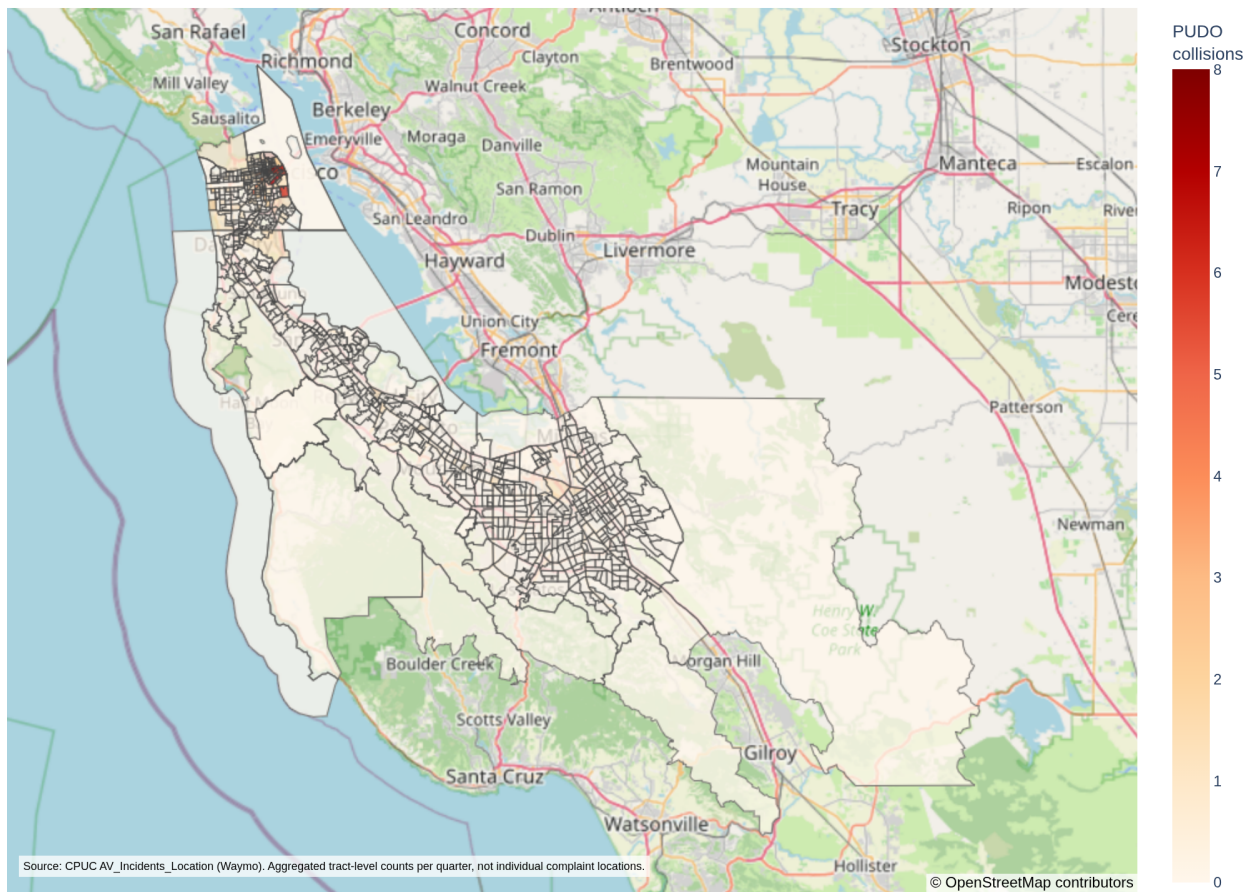


Figure 14: PUDO collisions by Census tract, SF Bay Area (2024 Q3 – 2026 Q2). The HTML edition of this paper has an interactive, pannable/zoomable version of this map.

The color scale is capped at 8 PUDO collisions rather than the true maximum of 18: collision counts per tract are heavily right-skewed (148 of 208 nonzero tracts have exactly one), so an uncapped linear scale would render almost every tract the same pale color. The two tracts above the cap (12 and 18) simply render at maximum intensity; their exact counts remain visible on hover in the HTML version and in Table 16 below.

San Francisco stands out: despite far fewer tracts than LA County (280 vs. 1,102), it accounts for the largest share of PUDO collisions (225 of 359, 63%), with the single densest concentration visible as the darkest tract in the SF Bay Area map, in San Francisco proper.

Table 16: Top 10 Census tracts by total PUDO collisions, summed across all available analysis quarters. Aggregated tract-level counts, not individual complaint locations.

tract GEOID	PUDO collisions	all collisions	periods reporting	periods
06075011700	55.0	8		2024Q3, 2024Q4, 2025Q1, 2025Q2, 2025Q3, 2025Q4, 2026Q1, 2026Q2
06075010100	21.0	6		2025Q1, 2025Q2, 2025Q3, 2025Q4, 2026Q1, 2026Q2
06075018000	27.0	8		2024Q3, 2024Q4, 2025Q1, 2025Q2, 2025Q3, 2025Q4, 2026Q1, 2026Q2
06075061500	20.0	6		2025Q1, 2025Q2, 2025Q3, 2025Q4, 2026Q1, 2026Q2
06075061500	15.0	6		2025Q1, 2025Q2, 2025Q3, 2025Q4, 2026Q1, 2026Q2
06075022600	8.0	8		2024Q3, 2024Q4, 2025Q1, 2025Q2, 2025Q3, 2025Q4, 2026Q1, 2026Q2
06075010600	21.0	8		2024Q3, 2024Q4, 2025Q1, 2025Q2, 2025Q3, 2025Q4, 2026Q1, 2026Q2
06075061500	8.0	2		2024Q3, 2024Q4
06075020500	7.0	8		2024Q3, 2024Q4, 2025Q1, 2025Q2, 2025Q3, 2025Q4, 2026Q1, 2026Q2
06075012100	10.0	8		2024Q3, 2024Q4, 2025Q1, 2025Q2, 2025Q3, 2025Q4, 2026Q1, 2026Q2

A second, narrower view adjusts for exposure — PUDO collisions per 100,000 trips — but only for **2024 Q3–Q4**: `AV_Monthly_Tract's TripsStart/ TripsEnd` are redacted at source for every period from 2025 Q1 onward, so a per-trip rate cannot be computed for the majority of the study window. It is a supplement to, not a replacement for, the full-window raw-count maps above.

Table 17: Top 10 Census tracts by PUDO-collision rate per 100k trips, 2024Q3 - 2024Q4 only (`TripsStart/TripsEnd` are redacted from 2025 Q1 onward, so this exposure-adjusted view cannot be extended to the rest of the study window).

tract GEOID	PUDO collisions	trips (start+end)	rate per 100k trips
06037191902	1.0	3332.0	30.01
06037700200	1.0	5055.0	19.78
06037701801	1.0	5805.0	17.23
06037214800	1.0	7404.0	13.51
06037700300	1.0	7444.0	13.43
06037190100	1.0	7752.0	12.9
06037190700	1.0	9134.0	10.95

tract GEOID	PUDO collisions	trips (start+end)	rate per 100k trips
06075021600	1.0	12006.0	8.33
06075017101	1.0	15051.0	6.64
06075021100	1.0	19533.0	5.12

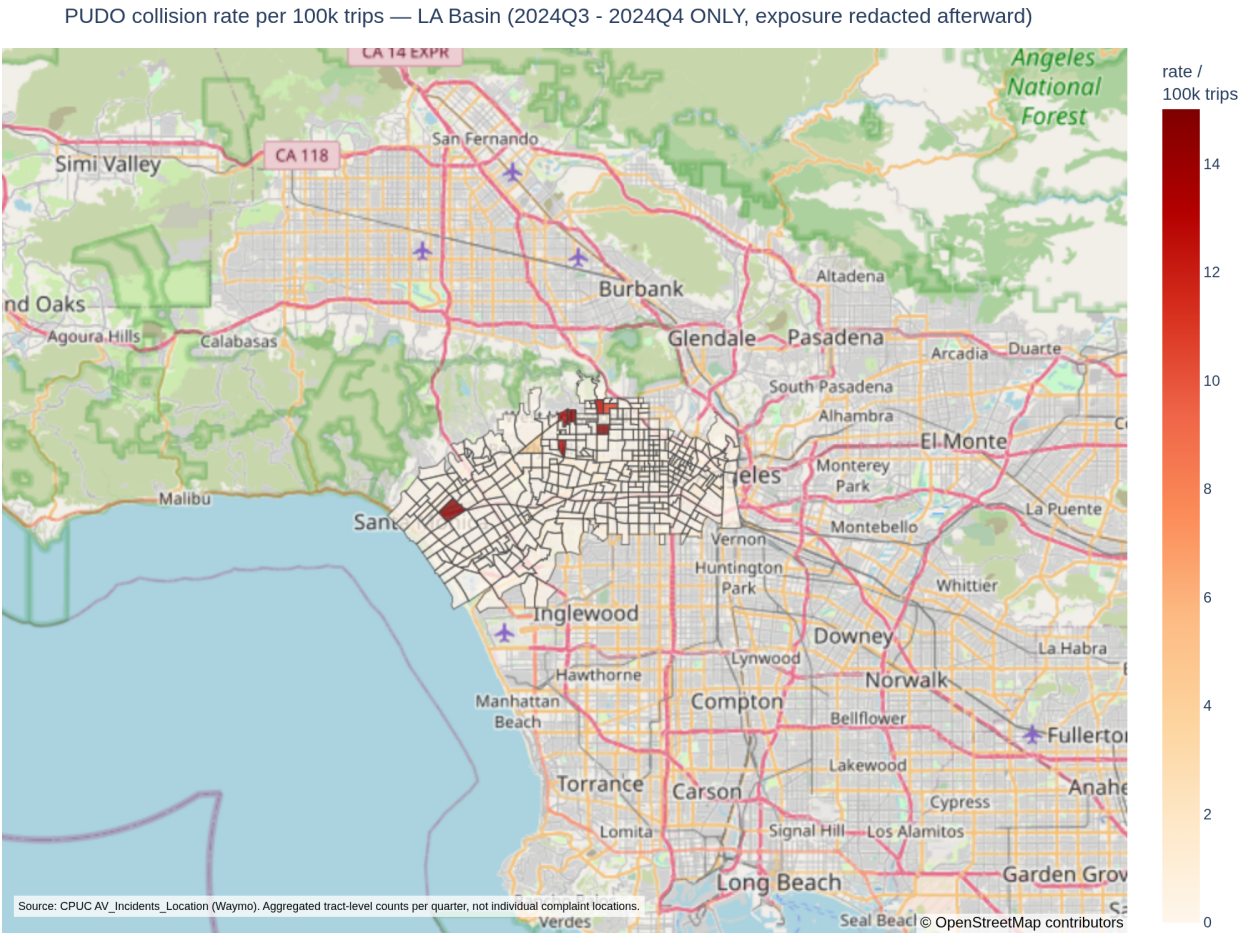


Figure 15: PUDO collision rate per 100,000 trips by tract, LA Basin — 2024 Q3–Q4 ONLY (exposure data are redacted for every later quarter).

PUDO collision rate per 100k trips — SF Bay Area (2024Q3 - 2024Q4 ONLY, exposure redacted afterward)

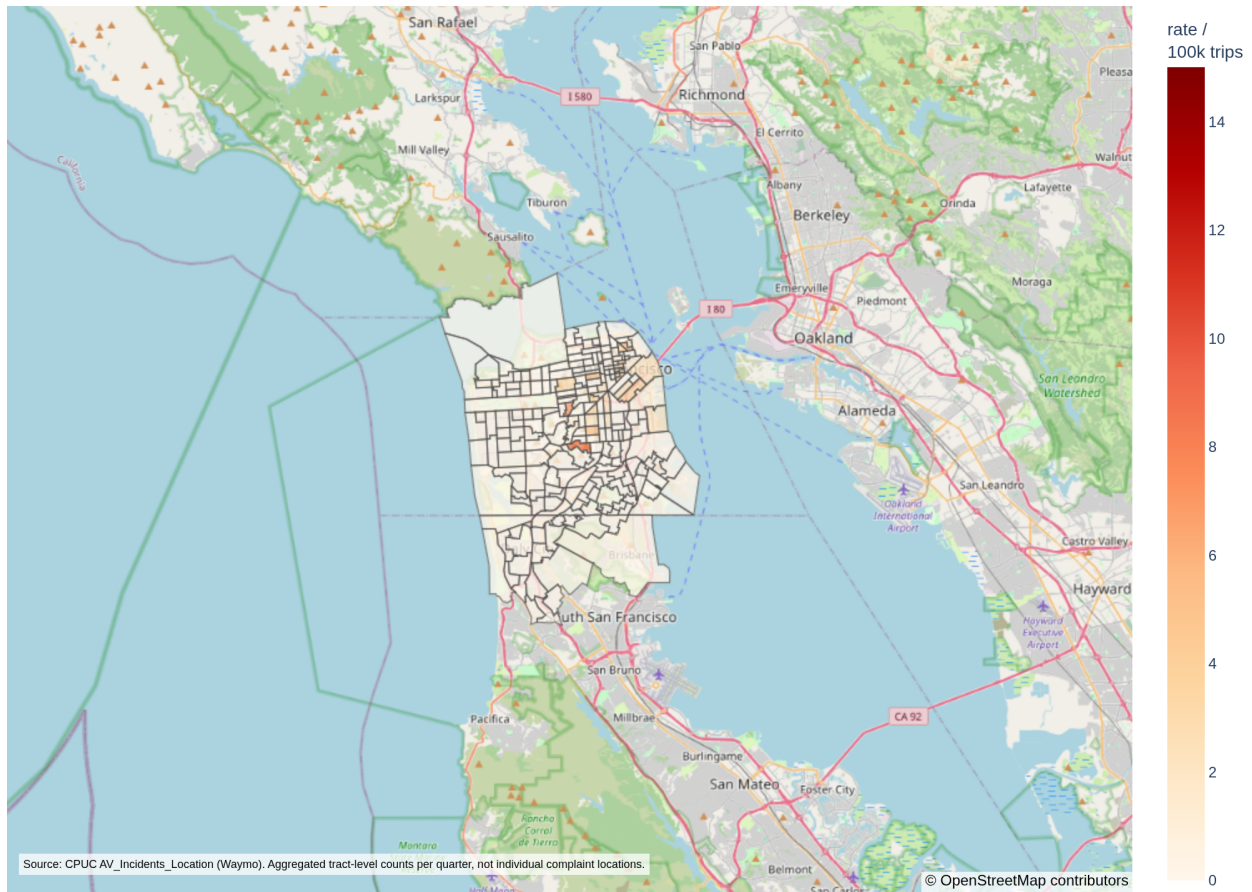


Figure 16: PUDO collision rate per 100,000 trips by tract, SF Bay Area — 2024 Q3–Q4 ONLY (exposure data are redacted for every later quarter).

6 Discussion

6.1 Key Concepts and Definitions

- **PUDO** — pick-up / drop-off: the maneuver where a vehicle stops to let a rider board or alight.
- **Exposure denominator** — the activity measure a complaint count is divided by (trips or VMT) so rates are comparable across quarters of very different fleet size.
- **Overdispersion** — when a count’s variance exceeds its mean beyond what a Poisson process allows, indicating real between-period rate variation.
- **Risk matrix axes** — *Probability*: how likely the hazard is to occur in a given period; *Impact*: how severe the outcome is when it does.
- **Credible interval vs. confidence interval** — the Bayesian model’s 95% credible interval is a direct probability statement (“there is a 95% chance the true slope is in this range, given the data and the prior”); the frequentist confidence interval is not, which is why the two are reported side by side rather than treated as interchangeable.

- **Descriptive vs. inferential interval** — the exact Poisson interval on each observed quarter (Equation 5) describes uncertainty around that quarter’s *count*; the Negative-Binomial regression’s confidence interval on β_1 describes uncertainty about the *trend across* quarters. They answer different questions, so overlapping per-quarter intervals are not evidence that no trend exists.

6.2 Assumptions

- PUDO complaints are initially treated as a Poisson-type count process; the overdispersion test shows this assumption fails, so Negative-Binomial regression — not Poisson — is used for all trend inference.
- The Bayesian trend model’s priors are weakly informative and were checked for sensitivity (Table 11) — the posterior does not depend meaningfully on the specific prior width chosen.
- The risk matrix’s “Almost certain” probability reading assumes the relevant question is *will at least one PUDO complaint occur somewhere in the fleet this quarter* (it will), not the odds for any single trip.
- Complaint counts are assumed complete and consistently defined within each reporting era; the 2024 aggregate table and the 2025+ ride-level microdata are assumed to measure the same underlying thing.

6.3 Limitations

- Complaints are self-reported by the carrier to its regulator; the rate measures *reported* problems, not necessarily underlying PUDO behavior, and reporting propensity could itself change over time.
- Waymo expanded to new cities during the window; a rate change may partly reflect new-market composition rather than a change in PUDO behavior.
- The 2024 and 2025+ numerators come from different source-file schemas, so a level shift at that boundary cannot be fully distinguished from a real trend.
- The launch-period sensitivity check further shrinks an already-small sample (post-ramp $n = 6$), which limits its power: a non-significant post-ramp slope is also consistent with a real but smaller decline continuing, not only with no decline at all.
- Redaction removes all within-quarter timing and location detail, and the travel-lane PUDO signal is unavailable in every file.
- Single carrier, single state, eight quarters — not yet a cross-carrier or multi-state comparison.
- The latest-quarter contrasts (*Is the latest quarter different?*) add a further small-sample constraint on top of the post-ramp window itself: with only $n = 6$ quarters and a 1-vs-5 split, they can rule out only a large shift in the most recent quarter, not a small one — the same caveat that applies to the post-ramp sensitivity check above.
- The potential software-release effect is a temporal association with a plausible engineering mechanism, not established causal evidence that the November–December 2024 software update changed PUDO behavior specifically.

7 Conclusion and Future Work

Over 2024 Q3 – 2026 Q2, the primary Negative-Binomial trend model finds Waymo’s exposure-adjusted PUDO complaint rate falling by 10.7% per quarter (95% CI -17.8% to -3.1% , $p = 0.007$) — not the rising trend a naive raw-count reading would suggest. The launch-period sensitivity analysis indicates that decline is driven substantially by the 2024 launch-ramp quarters: restricted to 2025 Q1 – 2026 Q2, the slope is small and not statistically significant. The Bayesian refit agrees in direction but is more cautious, putting the posterior probability of an actual decline at about 90% rather than treating the frequentist p -value as certainty. On a probability $\times$ impact risk matrix the hazard currently sits in the **Medium**, monitor-and-mitigate band: frequent enough to expect every quarter, but low-consequence, with no severe or fatal PUDO events observed. These results are still provisional (see Limitations) — in particular, the post-ramp window’s small sample cannot rule out a smaller ongoing decline, only a large one; as more CPUC quarters are published the same pipeline can extend the trend directly, and applying it to other CPUC-reporting AV carriers would turn a single-carrier trend into a cross-carrier comparison.

Three distinct eras run through this update: an elevated 2024 launch/ramp period; a comparatively stable 2025–2026 operating period that begins right around a documented Waymo Driver software release (deployed fleet-wide between November 7 and December 26, 2024) — a plausible engineering explanation for the launch-period drop, though not established as causal; and the question of whether the *latest available* quarter breaks that stability. On that last question, both regression-based contrasts (a regime-indicator model and a leave-latest-out predictive check, run because neither a rank test nor a t -test is valid at this sample size) and the matching Bayesian check find no statistically significant departure from the rest of the post-ramp window — the latest quarter reads as consistent with, not a break from, the 2025–2026 pattern, though the small post-ramp sample can only rule out a large shift, not a small one. Beyond this single deployment, PUDO-type complaint patterns are an early empirical data point for AV curbside risk as deployments expand to new cities and conditions — a signal worth continuing to track as new CPUC quarters arrive, not a basis for projecting the California rates found here onto other markets.

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